

ECE2202 Spring 2026 Exam 1 Review Packet Key

3.10 Apply nodal analysis to find node voltages V_1 to V_3 in the circuit of **Fig. P3.10** and then determine I_x .

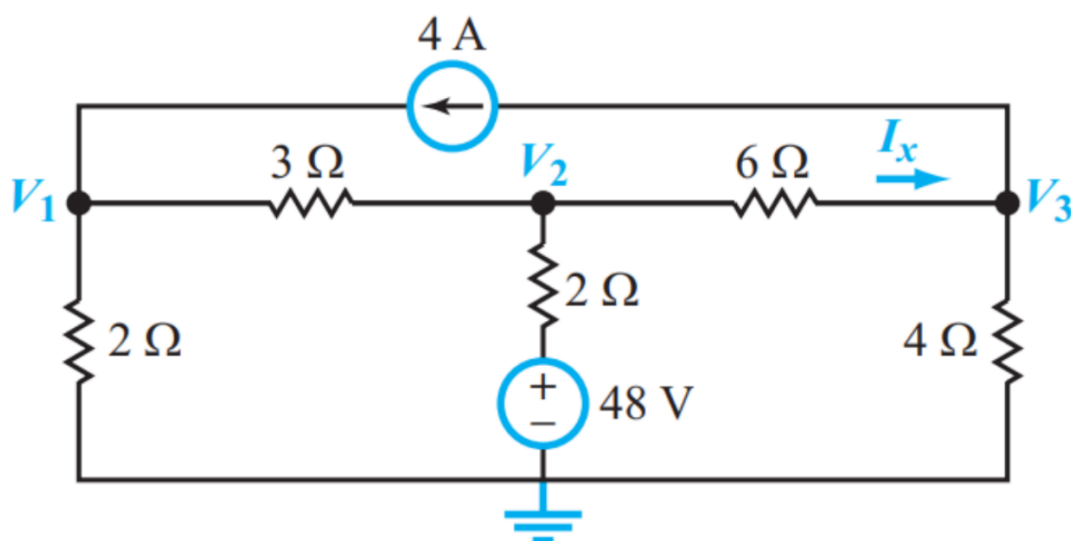


Figure P3.10 Circuit for Problem 3.10.

3.10

$$\text{NVM: KCL at } V_1: -\frac{V_1}{2} - \frac{(V_1 - V_2)}{3} + 4 = 0$$

$$\text{KCL at } V_2: -\frac{(V_2 - V_1)}{3} - \frac{(V_2 - 48)}{2} - \frac{(V_2 - V_3)}{6} = 0$$

$$\text{KCL at } V_3: -\frac{V_3 - V_2}{6} - \frac{V_3}{4} - 4 = 0$$

Solve for V_1, V_2, V_3

$$V_1 = 16.8[V], V_2 = 30[V], V_3 = 2.4[V]$$

$$\text{Ohm's Law: } I_x = \frac{V_2 - V_3}{6} = \frac{27.6}{6} = 4.6[A]$$

$$\boxed{I_x = 4.6[A]}$$

3.12 The magnitude of the dependent current source in the circuit of **Fig. P3.12** depends on the current I_x flowing through the $10\ \Omega$ resistor. Determine I_x .

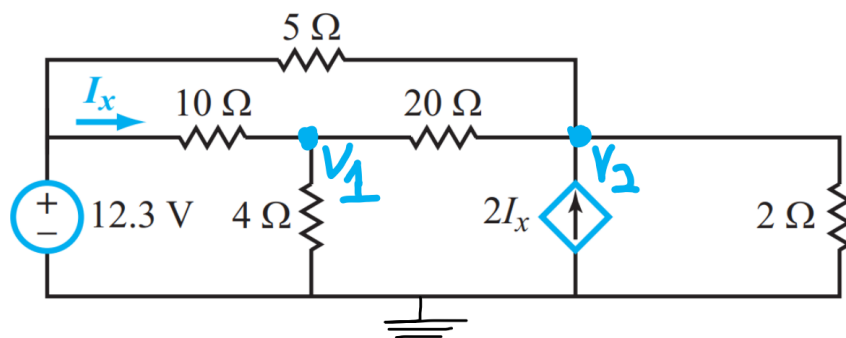


Figure P3.12 Circuit for Problems 3.12 and 3.13.

3.12 Ohm's Law: $I_x = \frac{12.3 - V_1}{10}$

NVM:
$$-\frac{V_1 - 12.3}{10} - \frac{(V_1 - V_2)}{20} - \frac{V_1}{4} = 0$$

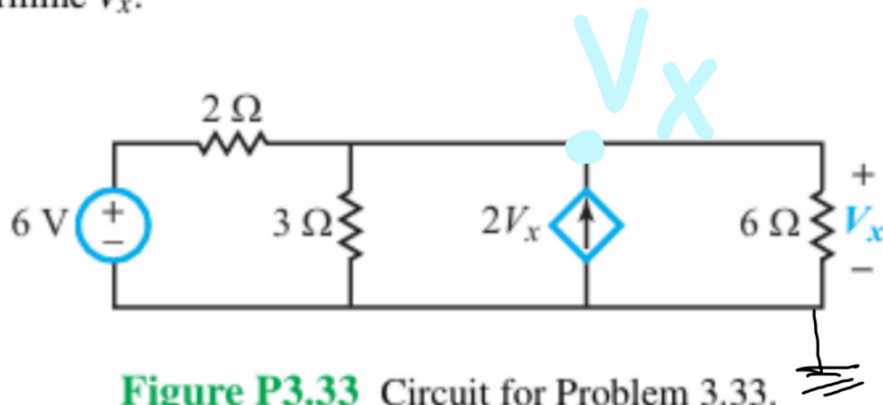
$$-\frac{(V_2 - V_1)}{20} + 2I_x - \frac{V_2}{2} - \left(\frac{V_2 - 12.3}{5}\right) = 0$$

Solve V_1, V_2, I_x

$$V_1 = 3.6[V], V_2 = 5.4[V], I_x = 0.45[A]$$

$I_x = 0.45[A]$

3.33 Apply mesh analysis to the circuit in **Fig. P3.33** to determine V_x .



3.33

For Node Voltage, V_x is the voltage of the one non-ground extraordinary node w/ respect to ground.

Node V_x :

$$\frac{V_x - 6}{2} + \frac{V_x}{3} - 2V_x + \frac{V_x}{6} = 0$$

Solve V_x :

$$V_x = -3[V]$$

3.34 Use the supermesh concept to solve for I_x in the circuit of **Fig. P3.34**.

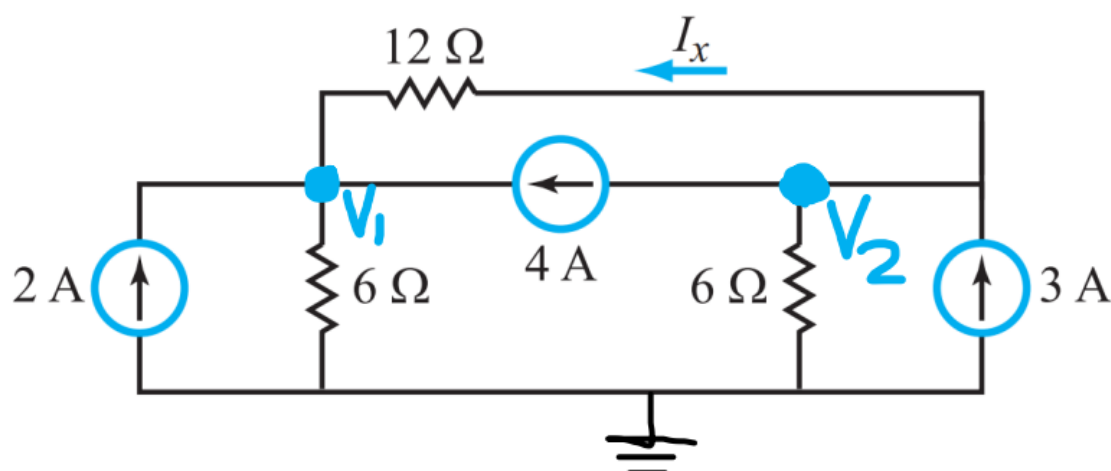


Figure P3.34 Circuit for Problem 3.34.

3.34

Node V_1 :

$$-2 + \frac{V_1}{6} - 4 + \frac{V_1 - V_2}{12} = 0$$

Node V_2 :

$$4 - 3 + \frac{V_2}{6} + \frac{V_2 - V_1}{12} = 0$$

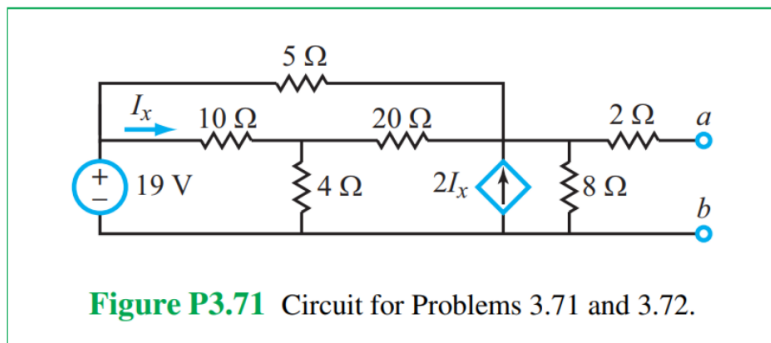
Solve V_1, V_2 $V_1 = 25.5 \text{ [V]}$ $V_2 = 4.5 \text{ [V]}$

$$I_x = \frac{V_2 - V_1}{12} = -\frac{21}{12} = -1.75 \text{ A}$$

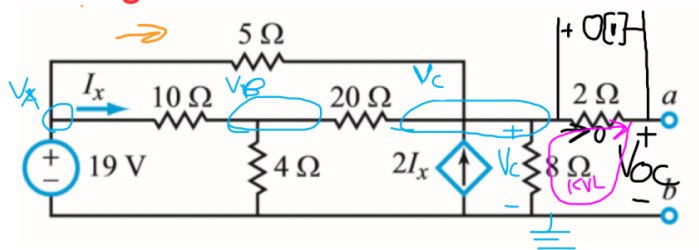
$$I_x = -1.75 \text{ [A]}$$

3.37 is the same as 3.12

3.71



Finding Voc:



$$V_A = 19[V]$$

$$\text{Node } V_B: \frac{V_B}{4} + \frac{V_B - 19}{10} + \frac{V_B - V_C}{20} = 0$$

$$\text{Node } V_C: \frac{V_B - V_C}{20} + 2I_x + \frac{19 - V_C}{5} - \frac{V_C}{8} = 0$$

$$I_x = \frac{19 - V_B}{10}$$

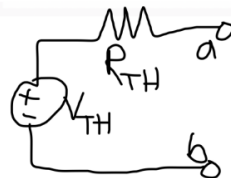
PLEASE NOTE THAT THE NUMERICAL VOLTAGE VALUES FROM THE ISC CIRCUIT CANNOT BE REUSED IN THE VOC CIRCUIT; THEY ARE TWO COMPLETELY DIFFERENT CIRCUITS!

$$V_C = 17.49[V], V_B = 6.94[V], I_x = 1.206[A]$$

$$\text{KVL: } V_{OC} - V_C + 0 = 0 \quad V_{OC} = V_C$$

$$V_{OC} = 17.49[V]$$

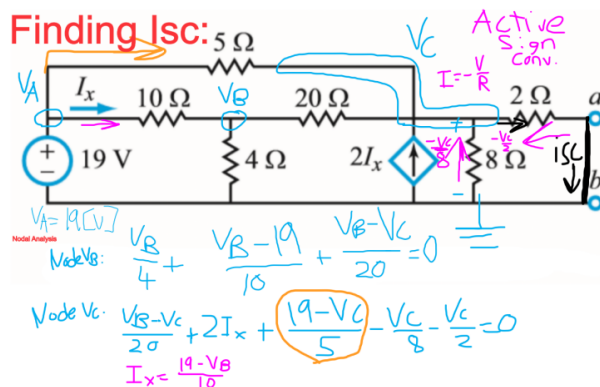
$$V_{TH} = V_{OC} \quad R_{TH} = R_{eq} = \frac{V_{OC}}{I_{SC}} = 4.53\Omega$$



For Thevenin, 3 options.

Find I_sc, V_oc, Req

Finding I_sc:



$$V_A = 19[V]$$

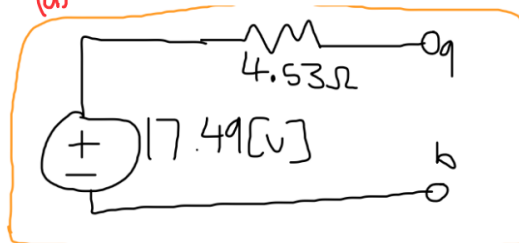
$$\text{Node } V_B: \frac{V_B}{4} + \frac{V_B - 19}{10} + \frac{V_B - V_C}{20} = 0$$

$$\text{Node } V_C: \frac{V_B - V_C}{20} + 2I_x + \frac{19 - V_C}{5} - \frac{V_C}{8} = 0$$

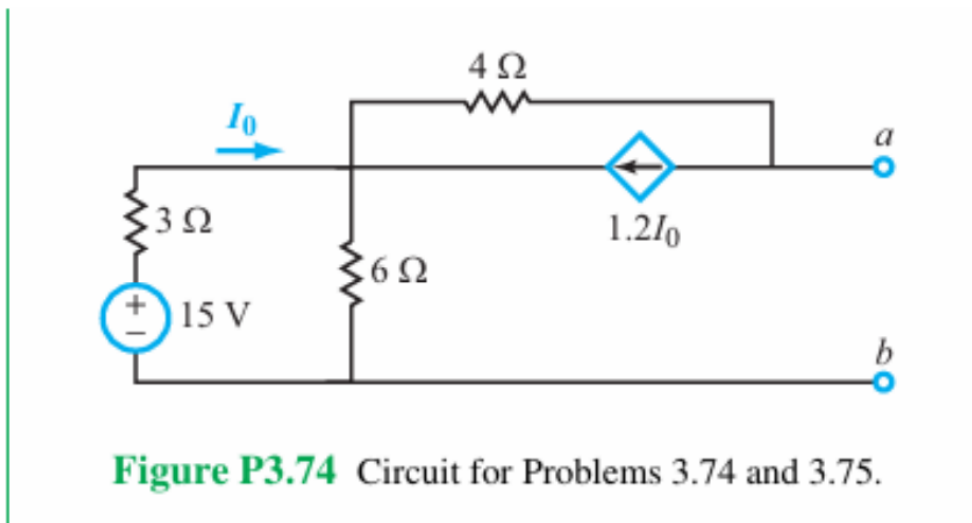
$$I_x = \frac{19 - V_B}{10}$$

$$I_{SC} = \frac{V_C}{2} = 3.8[A]$$

(a)

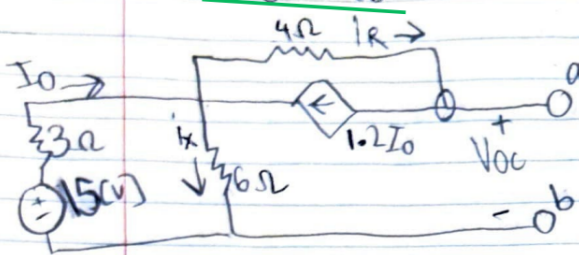


*3.74 Find the Norton equivalent circuit at terminals (a,b) of the circuit in Fig. P3.74.



3.74

Find V_{oc} :



$$KCL: i_R = 1.2I_0 + 0$$

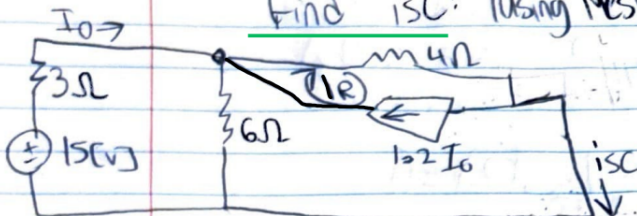
$$KCL: i_x = I_0 + 1.2I_0 - 1.2I_0 = I_0$$

$$KVL: V_{oc} - I_0 \cdot 6 + 1.2I_0 \cdot 4 = 0$$

$$KVL: 15 - I_0 \cdot 6 - I_0 \cdot 3 = 0$$

$$V_{oc} = 2[V]$$

Find i_{sc} : (using Mesh Current Method)



$$(i_{sc} - I_0) \cdot 6 + i_R \cdot 4 = 0$$

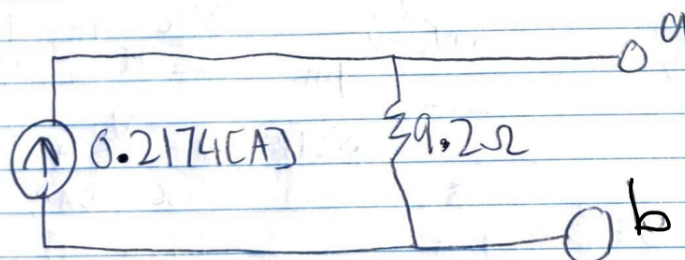
$$i_R - i_{sc} = 1.2I_0$$

$$I_0 \cdot 3 + (I_0 - i_{sc}) \cdot 6 - 15 = 0$$

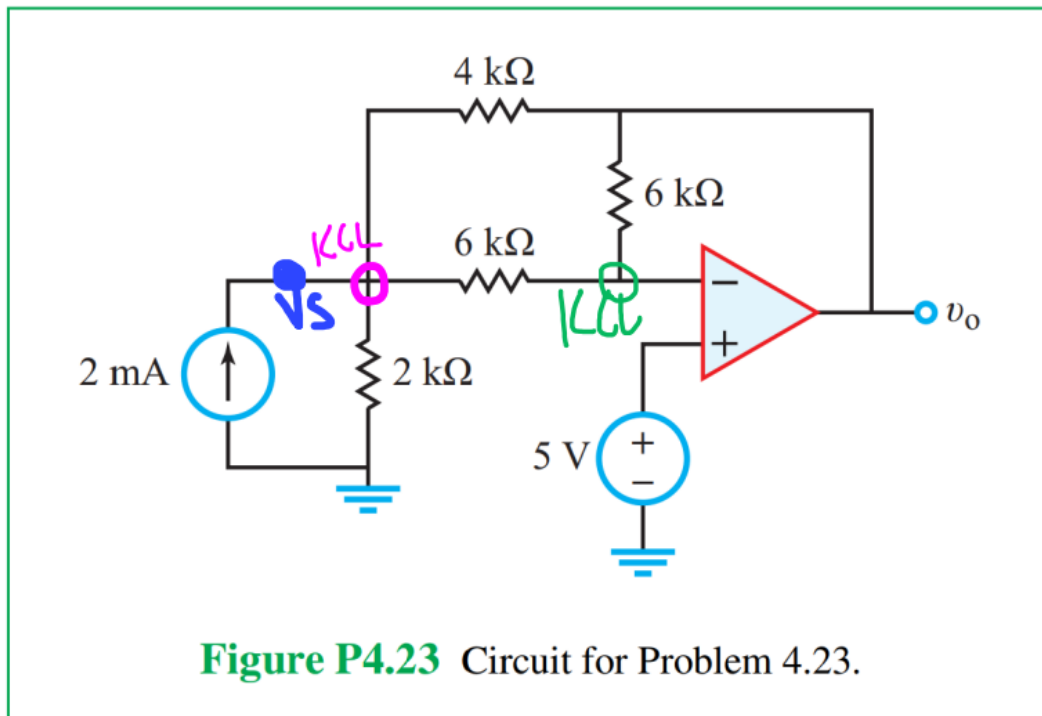
$$\text{Solve } i_{sc}: i_{sc} = 0.2174[A]$$

$$R_{eq} = \frac{V_{oc}}{i_{sc}} = 9.2\Omega$$

Norton Equivalent:



*4.23 Find the value of v_o in the circuit in Fig. P4.23.



$$\text{KCL: } \frac{5 - v_s}{6000} + \frac{5 - v_o}{6000} = 0$$

$$\text{KCL: } -0.002 + \frac{v_s}{2000} + \frac{v_s - 5}{6000} + \frac{v_s - v_o}{4000} = 0$$

2 equations, 2 unknowns; Solve!

$$v_o = 5.42857 \text{ [V]}, v_s = 4.5714 \text{ [V]}$$

4.24 For the circuit in Fig. P4.24, obtain an expression for voltage gain $G = v_o/v_s$.

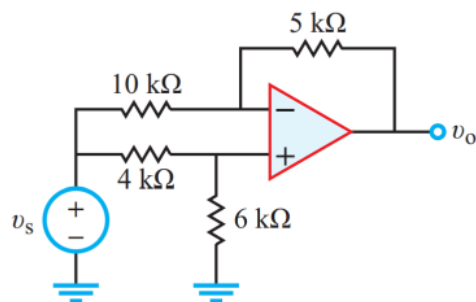
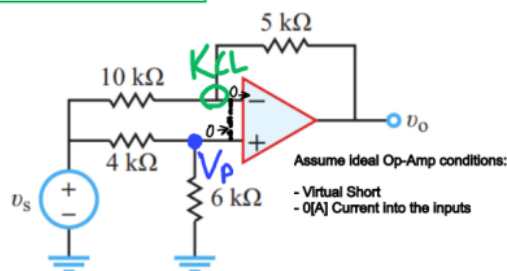


Figure P4.24 Circuit for Problem 4.24.



4 [kΩ] & 6 [kΩ] are technically in series,
 Since 0[A] current goes into the + input.
 Therefore, we may use Voltage Divider to get V_p
 in terms of V_s .

$$V_p = \left(\frac{6000}{6000+4000} \right) \cdot V_s = \frac{3}{5} V_s, \quad V_p = \frac{3}{5} V_s$$

Green KCL: $\frac{\frac{3}{5} V_s - V_s}{10000} + \frac{\frac{3}{5} V_s - V_o}{5000} = 0$

Isolate $\frac{V_o}{V_s}$: $\downarrow \times 10000$

$$\frac{3}{5} V_s - V_s + 2 \left(\frac{3}{5} V_s - V_o \right) = 0$$

$$-\frac{2}{5} V_s + \frac{6}{5} V_s - 2 V_o = 0$$

$$\frac{4}{5} V_s = 2 V_o$$

$$\boxed{\frac{V_o}{V_s} = \frac{4}{10} = 0.4}$$